

EE 445/645: Physical Models in Remote Sensing (Spring 2026)

Chapter 02-Part 02: The Extinction Coefficient – Notes (See C2-P2 PPTs)

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§1. Introduction.

In this chapter we shall derive the extinction coefficient for a vegetation canopy idealized as a turbid medium. The extinction coefficient is the fundamental quantity that governs the attenuation of radiation as it propagates through the canopy; its determination is therefore prerequisite to the solution of any radiative transfer problem in vegetated media.

We shall find that the extinction coefficient admits a natural factorization into two quantities of distinct physical character: the leaf area density, which describes the amount of foliage per unit volume, and the *geometry function* G , which encodes the angular distribution of leaf orientations. The geometry function, first introduced by Ross (1981), is remarkable in that it renders the extinction coefficient dependent upon the direction of photon travel but independent of wavelength—a property that distinguishes radiative transfer in vegetation from that in all other natural media.

After deriving the extinction coefficient and establishing the normalization of the G -function, we shall examine its properties, discuss the limitations of the turbid-medium formulation, present explicit analytical forms for several classical leaf-angle distributions, and treat the special case of coniferous canopies in which needles are clustered into shoots. Complete derivations of all results quoted in §8 are given in the Appendix.

§2. The turbid-medium abstraction.

Canopies of natural vegetation exhibit formidable complexity. Branches twist and turn in every direction, leaves overlap and cluster at irregular intervals, and the architecture changes from season to season. Rather than attempt to capture this complexity in all its detail, we resort to a simple abstraction: the canopy is idealized as a medium densely filled with small planar elements of negligible thickness and area. This representation is referred to as a turbid medium, or, colloquially, a green gas (Fig. 1).

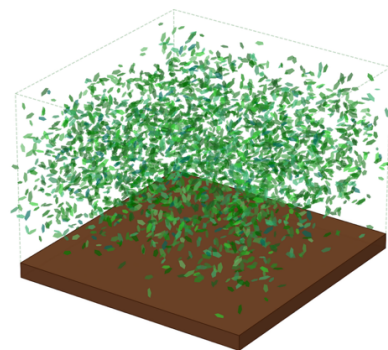


Figure 1. Vegetation canopy idealized as a turbid of green planar elements.

The abstraction rests upon the following assumptions. First, the canopy medium is filled with a sufficiently large number of small planar elements that macroscopic variables such as the leaf area density and the leaf normal orientation distribution may be treated as continuous and differentiable functions of position. Second, each element is oriented in space according to a probability density function $g_L(\mathbf{r}, \mathbf{\Omega}_L)$ defined over the hemisphere of leaf-normal directions $\mathbf{\Omega}_L$. Third, and most critically, the projections of these elements onto any plane perpendicular to an arbitrary direction $\mathbf{\Omega}$ do not overlap. It is this no-overlap assumption that permits us to compute the total shadow area as the arithmetic sum of the individual shadow areas, and it forms the foundation of the entire derivation that follows.

§3. Definition of the extinction coefficient.

The *extinction coefficient* (total interaction cross section) is defined in terms of the probability that a photon, while traveling an infinitesimal distance $d\xi$ through the medium, will interact with the elements of the host medium. This probability is given by the product

$$\sigma(\mathbf{r}, \mathbf{\Omega}) d\xi, \quad (1)$$

where $\sigma(\mathbf{r}, \mathbf{\Omega})$ is the extinction coefficient at position $\mathbf{r}$ for the direction of photon travel $\mathbf{\Omega}$. It has dimensions of inverse length (m^{-1}).

The quantity σ thus measures, per unit distance traveled, the likelihood that a photon will be intercepted by the foliage. A large value of σ indicates an optically dense medium in which photons are rapidly absorbed or scattered; a small value indicates a relatively transparent medium. We note that σ depends upon both position, because the canopy may be denser in some regions than in others, and upon the direction of photon travel, because the projected area of the leaves varies with the direction from which they are viewed.

§4. Derivation of the extinction coefficient.

Consider an elementary volume $\Delta V = dS d\xi$ at position $\mathbf{r}$ in the canopy. This small cylinder has cross-sectional area dS perpendicular to the direction of photon travel $\mathbf{\Omega}$ and depth $d\xi$ along $\mathbf{\Omega}$. We suppose that the volume contains a sufficient number of small planar leaf elements of negligible thickness.

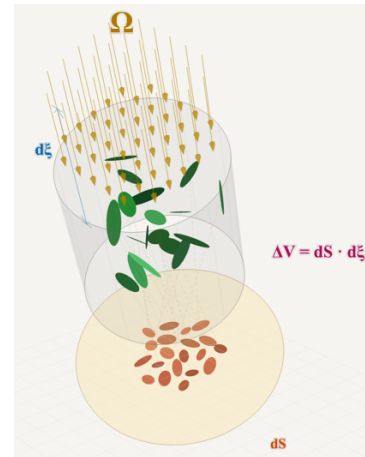


Figure 2. Projection of leaf elements in an elementary volume ΔV at $\mathbf{r}$ on to a plane perpendicular to the direction of photon travel $\mathbf{\Omega}$.

The probability that a photon comprising the incident beam will collide with a leaf element in this volume is equal to the ratio (Fig. 2)

$$\sigma(\mathbf{r}, \boldsymbol{\Omega}) d\xi = \frac{\text{total shadow area of leaves on a plane perpendicular to } \boldsymbol{\Omega}}{dS}.$$

We now express the total shadow area explicitly. Let the volume contain N leaf elements, the i th of which has one-sided area s_i and leaf-normal direction $\boldsymbol{\Omega}_{Li}$. The shadow (projected area) of this element on a plane perpendicular to $\boldsymbol{\Omega}$ is $s_i |\boldsymbol{\Omega}_{Li} \cdot \boldsymbol{\Omega}|$. Since, by assumption, the individual shadows do not overlap, the total shadow area is the arithmetic sum

$$\sigma(\mathbf{r}, \boldsymbol{\Omega}) d\xi = \frac{\sum_{i=1}^N s_i |\boldsymbol{\Omega}_{Li} \cdot \boldsymbol{\Omega}|}{dS}.$$

Counting over individual elements is tedious when N is large. It is more convenient to group the elements by the direction of their normals (Fig. 3).

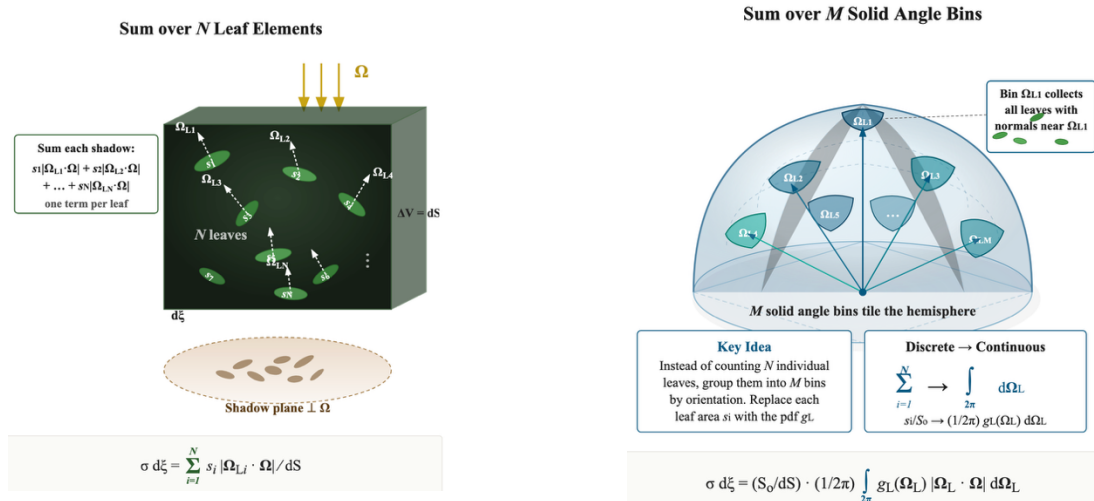


Figure 3. Two equivalent ways to compute the extinction cross-section σ of a vegetation canopy modeled as a turbid medium. Left panel shows the direct approach: each of the N leaf elements (areas $s_1, s_2, \dots, s_N$) in the elementary volume is individually projected onto a plane perpendicular to the photon travel direction $\boldsymbol{\Omega}$, and their shadow areas are summed term by term. The projection of leaf i with normal $\boldsymbol{\Omega}_{Li}$ contributes $s_i |\boldsymbol{\Omega}_{Li} \cdot \boldsymbol{\Omega}|$ to the total shadow. This is exact but impractical when N is large. Right panel shows the efficient alternative: instead of enumerating leaves one by one, the upper hemisphere of leaf normal directions is partitioned into M solid angle bins, each collecting all leaves whose normals fall within that bin. The fractional area of leaves in bin $\boldsymbol{\Omega}_{Li}$ is expressed through the leaf angle probability density $g_L(\boldsymbol{\Omega}_L)$, so the discrete sum over N leaves becomes a continuous integral over the hemisphere.

Let the hemisphere of leaf-normal directions be partitioned into M solid-angle increments $d\boldsymbol{\Omega}_{Li}$, $i = 1, \dots, M$. The ratio of the total one-sided area of all elements whose normals fall within $d\boldsymbol{\Omega}_{Li}$ to the total leaf area S_0 in the volume is, by definition, the probability of finding leaf elements of that orientation:

$$\frac{s_i(\boldsymbol{\Omega}_{Li})}{S_0} = \frac{1}{2\pi} g_L(\mathbf{r}, \boldsymbol{\Omega}_{Li}) d\boldsymbol{\Omega}_{Li},$$

whence

$$s_i(\mathbf{\Omega}_{Li}) = \frac{S_o}{2\pi} g_L(\mathbf{r}, \mathbf{\Omega}_{Li}) d\mathbf{\Omega}_{Li}.$$

Here g_L is the probability density function of leaf-normal orientation. Substituting into the sum and passing to the limit as $M \rightarrow \infty$, we obtain

$$\sigma(\mathbf{r}, \mathbf{\Omega}) d\xi = \frac{S_o}{dS} \left[\frac{1}{2\pi} \int_0^{2\pi} g_L(\mathbf{r}, \mathbf{\Omega}_L) |\mathbf{\Omega}_L \cdot \mathbf{\Omega}| d\mathbf{\Omega}_L \right].$$

We now identify the two factors on the right-hand side. The ratio $S_o/(dS d\xi)$ is the total one-sided leaf area per unit volume, which we denote by $u_L(\mathbf{r})$ and call the leaf area density (m^{-1}). The bracketed integral depends only upon the leaf-normal distribution and the direction of photon travel; we denote it by $G(\mathbf{r}, \mathbf{\Omega})$ and call it the geometry function, or G -function. Thus

$$\sigma(\mathbf{r}, \mathbf{\Omega}) = u_L(\mathbf{r}) G(\mathbf{r}, \mathbf{\Omega}), \quad (2)$$

where

$$G(\mathbf{r}, \mathbf{\Omega}) = \frac{1}{2\pi} \int_0^{2\pi} g_L(\mathbf{r}, \mathbf{\Omega}_L) |\mathbf{\Omega}_L \cdot \mathbf{\Omega}| d\mathbf{\Omega}_L. \quad (3)$$

This factorization is of fundamental importance: it cleanly separates the quantity of foliage, u_L , from the geometric arrangement of that foliage, G .

§5. The normalization of the G -function.

The G -function, also called the geometry factor, was first introduced by Juhan Ross in his classical treatise *The Radiation Regime and Architecture of Plant Stands* (1981). It satisfies an important normalization condition which we now derive.

Averaging the G -function over all directions of photon travel $\mathbf{\Omega}$ uniformly distributed over the hemisphere of 2π steradians, we obtain

$$\frac{1}{2\pi} \int_0^{2\pi} G(\mathbf{r}, \mathbf{\Omega}) d\mathbf{\Omega} = \frac{1}{2}. \quad (4)$$

The proof proceeds by substituting the definition of G and interchanging the order of integration. Writing $\mathbf{\Omega}_L \cdot \mathbf{\Omega} = \cos\theta$, where θ is the angle between the leaf normal and the direction of photon travel, the inner integral over all directions $\mathbf{\Omega}$ of the quantity $|\cos\theta|$ over the full sphere of 4π steradians yields 2π . The outer integral over $\mathbf{\Omega}_L$ then reduces to the normalization of g_L itself (Eq. (A.3)). A rigorous proof is given in Appendix C, §12–§14, where we establish the master normalization identity (Eq. (C.1)) $\int_0^{\pi/2} \psi(\mu, \mu_L) \sin\theta d\theta = 1/2$ and show that the normalization of G reduces to the normalization of $\bar{g}_L$.

The physical content of this result is noteworthy: the average projected area, when averaged uniformly over all possible directions of illumination, is exactly one-half of the total one-sided leaf area. This holds regardless of the leaf orientation distribution—whether the leaves are predominantly horizontal, predominantly vertical, or distributed in any other fashion.

§6. Properties of the G -function and the extinction coefficient.

The G -function and the extinction coefficient σ exhibit several properties that distinguish radiative transfer in vegetation canopies from that in other natural media.

Directional dependence. Both G and σ depend explicitly upon the direction of photon travel Ω . The single exception occurs when the leaf-normal distribution is uniform, $g_L(\mathbf{r}, \Omega_L) = 1$, in which case $G = 1/2$ independent of direction (see Appendix E, Problem 2, for the derivation).

Wavelength independence. The extinction coefficient is independent of photon frequency (wavelength), because the leaves and other organs of a vegetation canopy are much larger than the wavelengths of solar radiation. Extinction is a purely geometric phenomenon: it depends upon shapes and orientations, not upon wavelength. This property is the converse of that encountered in atmospheric and oceanic media, where the extinction coefficient is typically wavelength-dependent but direction-independent.

Structural dependence. It follows that only the canopy structure—the spatial distribution and orientation of foliage elements—determines the extinction of radiation in vegetation canopies. Radiative transfer in vegetation is therefore fundamentally unlike standard radiative transfer in atmospheres, water bodies, snowpacks, and similar media.

Inapplicability of classical methods. Because of the directional dependence of σ , the standard methods of solving the radiative transfer equation developed in astrophysics and atmospheric physics—which assume a direction-independent extinction coefficient—cannot in general be applied to radiative transfer in vegetation media.

Observational consequence. The extinction coefficient $\sigma(\mathbf{r}, \Omega)$ can be estimated from transmission measurements made below the canopy at wavelengths where leaves strongly absorb the incident radiation. Such measurements can, in principle, be inverted to recover the leaf area density $u_L(\mathbf{r})$ and the leaf-normal orientation distribution $g_L(\mathbf{r}, \Omega_L)$.

§7. Limitations.

The derivation of the extinction coefficient presented above rests upon assumptions that are, strictly speaking, contradictory.

On the one hand, we require that the elementary volume contain a sufficient number of leaf elements for the leaf area density and the leaf-normal distribution to be realized as continuous and differentiable functions. On the other hand, we assume that the projections of these elements do not overlap. If the volume contains enough elements to justify a continuous description, it becomes increasingly improbable that their shadows—which have finite extent—would not overlap.

In reality, there is considerable mutual shading between leaves in a canopy. Our formulation, being probabilistic, does not account for the finite size of leaves and the consequent overlap of their shadows. As we shall see in a later chapter, the finite dimensions of leaves give rise to correlations in photon mean free paths between directions close to one another, producing a brightening of reflectance about the retro-solar direction known as the hot spot. The present formulation does not capture this correlation, and the hot-spot effect is therefore not represented.

These limitations notwithstanding, the turbid-medium model has proved remarkably successful in a wide range of applications, and the extinction coefficient as derived here remains the standard starting point for vegetation radiative transfer theory.

§8. Analytical forms of the G -function.

Explicit analytical expressions for the G -function can be obtained for the classical leaf-normal inclination distributions. First, we consider the case when the azimuthal distribution of leaf normals is uniform, so that $h_L(\varphi_L) = 1$ and the G -function depends only upon the beam zenith angle θ . The extension to non-uniform azimuthal distributions is treated at the end of this section and derived in full in Appendix B.

The azimuthal projection integral. As shown in Appendix B, §7–§9, the G -function for a canopy with azimuthally uniform leaf normals may be written as

$$G(\theta) = \int_0^{\pi/2} \bar{g}_L(\theta_L) \psi(\mu, \mu_L) \sin\theta_L d\theta_L, \quad (5)$$

where $\mu = \cos\theta$, $\mu_L = \cos\theta_L$, and $\psi(\mu, \mu_L)$ is the azimuthal projection integral—the average of the absolute direction cosine $|\mathbf{\Omega}_L \cdot \mathbf{\Omega}|$ over the leaf azimuth. The leaf-normal inclination distribution $\bar{g}_L(\theta_L)$ is defined as a probability density function with respect to the solid-angle measure $\sin\theta_L d\theta_L$ on the upper hemisphere, so that

$$\int_0^{\pi/2} \bar{g}_L(\theta_L) \sin\theta_L d\theta_L = 1.$$

The evaluation of ψ yields the piecewise result (Appendix B, §9):

$$\psi(\mu, \mu_L) = \begin{cases} |\mu\mu_L|, & \text{if } |\mu\mu_L| \geq \sin\theta \sin\theta_L, \\ \mu\mu_L \left(\frac{2\varphi_t}{\pi} - 1 \right) + \frac{2}{\pi} \sin\theta \sin\theta_L \sin\varphi_t, & \text{otherwise,} \end{cases} \quad (6)$$

where $\varphi_t = \arccos(-\mu\mu_L/\sin\theta \sin\theta_L)$.

Special cases: single leaf inclination. When all leaves share a common inclination θ_0 , the distribution reduces to $\bar{g}_L(\theta_L) = \delta(\theta_L - \theta_0)/\sin\theta_0$, and the sifting property of the delta function gives $G(\theta) = \psi(\mu, \cos\theta_0)$ directly (Appendix E, Problem 1). For horizontal leaves ($\theta_0 = 0$) we recover $G = \cos\theta$ (Appendix E, Problem 1, Sub-case: Horizontal Leaves), and for vertical leaves ($\theta_0 = \pi/2$) we obtain $G = (2/\pi)\sin\theta$ (Appendix E, Problem 1, Sub-case: Vertical Leaves).

The standard trigonometric distributions. For general distributions, the evaluation proceeds by expressing $\bar{g}_L$ as a linear combination of 1, $\cos 2\theta_L$, and $\cos 4\theta_L$ (via the standard double-angle identities) and exploiting the linearity of the integral. As established in Appendix D, §15–§16, the G -function then decomposes as $G = c_0 \mathcal{J}_0 + c_2 \mathcal{J}_2 + c_4 \mathcal{J}_4$, where the fundamental integrals take the closed forms

$$\mathcal{J}_0 = \frac{1}{2}, \quad \mathcal{J}_2 = -\frac{\sin^2\theta}{4}, \quad \mathcal{J}_4 = -\frac{\cos^4\theta}{6}. \quad (7)$$

Substituting the expansion coefficients for each distribution (Appendix E) yields the following results plotted in Fig. 4.

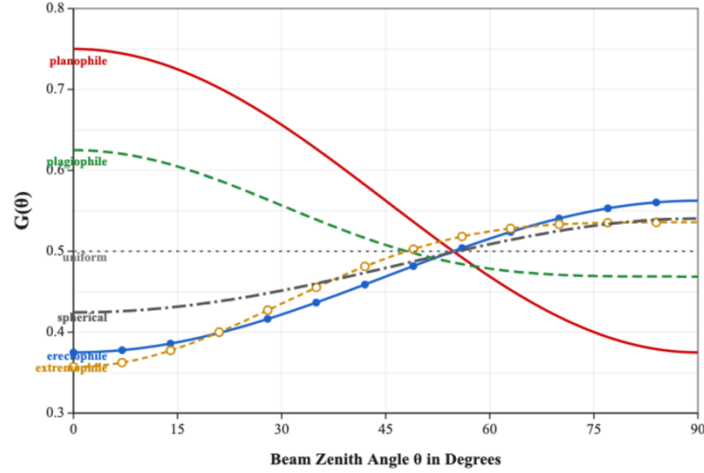


Figure 4. The geometry function G for various models of leaf normal orientation distributions.

Planophile ($\bar{g}_L = 3\cos^2\theta_L$; predominantly horizontal leaves):

$$G(\theta) = \frac{3(1+\cos^2\theta)}{8}. \quad (8)$$

This function is large for small θ (near-vertical illumination) and decreases as the beam becomes more oblique, reflecting the fact that horizontal leaves present their maximum projected area to a vertical beam.

Erectophile ($\bar{g}_L = \frac{3}{2}\sin^2\theta_L$; predominantly vertical leaves):

$$G(\theta) = \frac{3(2+\sin^2\theta)}{16}. \quad (9)$$

The behaviour is the converse of the planophile case: G is smaller for near-vertical beams and increases toward horizontal incidence.

Plagiophile ($\bar{g}_L = \frac{15}{8}\sin^2 2\theta_L$; leaves concentrated near 45°):

$$G(\theta) = \frac{5(3+\cos^4\theta)}{32}. \quad (10)$$

The G -function varies modestly with beam zenith angle, reflecting the intermediate character of this distribution.

Extremophile ($\bar{g}_L = \frac{15}{7}\cos^2 2\theta_L$; bimodal—horizontal and vertical):

$$G(\theta) = \frac{5(3-\cos^4\theta)}{28}. \quad (11)$$

Uniform distribution ($\bar{g}_L = 1$; all inclination angles equally probable):

$$G = \frac{1}{2}, \quad (12)$$

independent of θ , consistent with the result of Appendix E, Problem 2.

Each of these G -functions (Fig. 4) has been verified to satisfy the normalization condition $\int_0^{\pi/2} G(\theta) \sin\theta d\theta = 1/2$, as proved in general by the master identity of Appendix C, §12–§13 and confirmed case by case in Appendix E. The summary table of Appendix F, §18 collects all results.

Non-uniform azimuthal distributions: When the leaf azimuthal distribution is not uniform but takes the form $h_L(\varphi_L) = 2\cos^2(\varphi - \varphi_L - \eta)$, the azimuthal projection integral acquires an additional correction term. As derived in Appendix B, §8–§10, the result is

$$\psi_{NU}(\mu, \mu_L, \eta) = \frac{1}{2} \psi_U(\mu, \mu_L) + \Delta\psi(\mu, \mu_L, \eta), \quad (13)$$

where ψ_U is the uniform result and the correction $\Delta\psi$ vanishes whenever $|\mu\mu_L| \geq \sin\theta \sin\theta_L$ (Eq. (B.7) of Appendix B, §10). The non-uniform azimuthal preference thus makes no difference when the sun is nearly vertical or the leaves nearly horizontal; the correction is largest when both θ and θ_L are far from zero and the canopy is illuminated obliquely (Appendix B, §10).

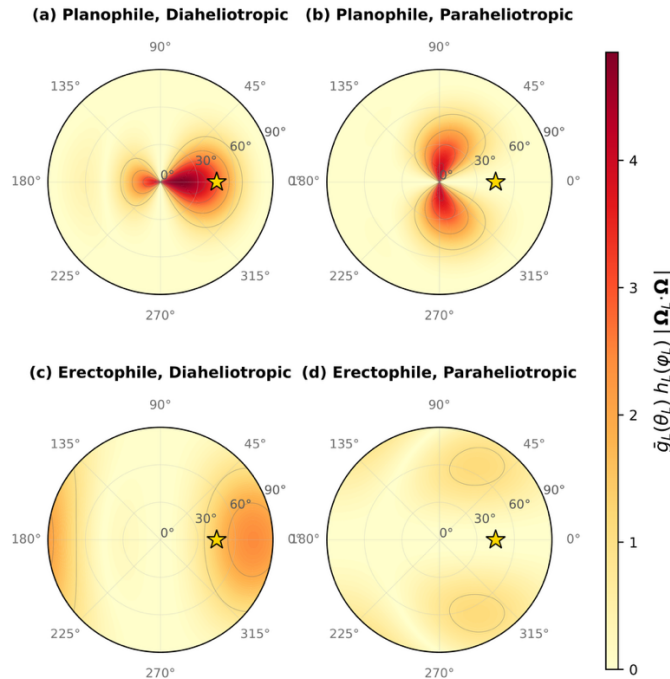


Figure 5. Projected leaf-normal density $\bar{g}_L(\theta_L) h_L(\varphi_L) |\mathbf{\Omega}_L \cdot \mathbf{\Omega}|$ on the upper hemisphere of leaf-normal directions for four combinations of inclination model $\bar{g}_L$ and azimuthal model h_L . The gold star marks the fixed beam direction ($\theta = 45^\circ$, $\varphi = 0^\circ$).

The interplay between the inclination distribution $\bar{g}_L(\theta_L)$ and the azimuthal distribution $h_L(\varphi_L)$ is illustrated in Figs. 5 and 6 for two representative inclination models — planophile and erectophile — crossed with the two extreme azimuthal preferences — diaheliotropic ($\eta = 0$) and paraheliotropic ($\eta = \pi/2$).

Figure 5 displays the *integrand* of the G -function, namely $\bar{g}_L(\theta_L) h_L(\varphi_L) |\mathbf{\Omega}_L \cdot \mathbf{\Omega}|$, on the upper hemisphere of leaf-normal directions (θ_L, φ_L) for a fixed beam at $\theta = 45^\circ$ and $\varphi = 0^\circ$. Fig. 5a

(planophile, diaheliotropic) concentrates the projected density near the center of the hemisphere, where nearly vertical normals belong to the predominantly horizontal leaves that face the incoming beam. Fig. 5b (planophile, paraheliotropic) redirects the density into two lobes at $\varphi_L = 90^\circ$ and 270° , because the leaves now preferentially orient their normals perpendicular to the beam azimuth, presenting less projected area. Fig. 5c (erectophile, diaheliotropic) shifts the density toward the rim, reflecting the predominantly vertical leaf orientations that project most effectively when their normals are aligned with the oblique beam. Fig. 5d (erectophile, paraheliotropic) is the weakest of the four: vertical leaves turned away from the beam intercept very little radiation at any leaf-normal direction.

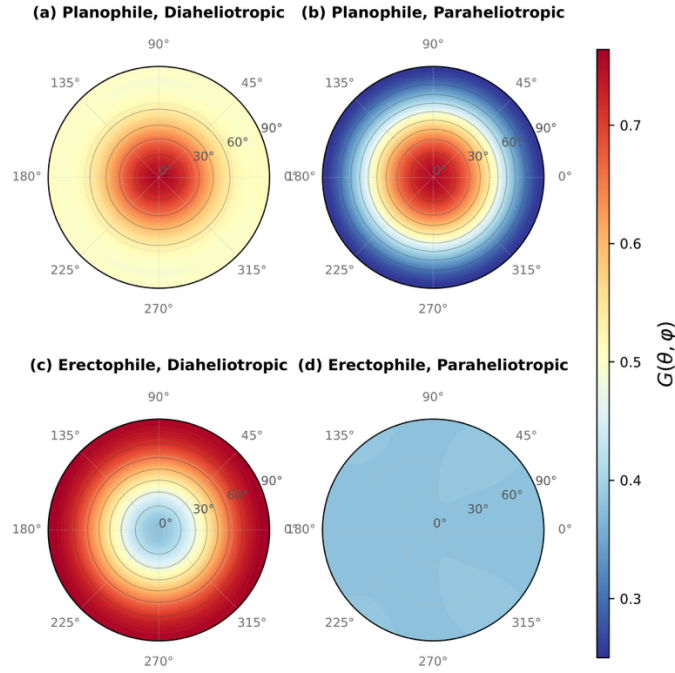


Figure 6. The geometry function $G(\theta, \varphi)$ evaluated for beam directions $\mathbf{\Omega} = (\theta, \varphi)$ on the upper hemisphere, for the same four model combinations as Fig. 5.

Figure 6 shows the result of integrating these densities: the G -function itself, plotted over all beam directions (θ, φ) . The radial coordinate is the beam zenith angle. Fig. 6a confirms that planophile, diaheliotropic canopies yield the highest G for near-vertical beams, as expected from the concentration of projected density at the center of Fig. 5a. Fig. 5b shows the paraheliotropic correction at work: G drops sharply for overhead beams (where the avoidance is most effective) but remains high at oblique angles where the azimuthal preference matters less. Fig. 5c exhibits the opposite zenith dependence — G is low overhead and rises steeply toward the horizon — consistent with the rim-concentrated density of Fig. 5c. Fig. 6d is the most striking: the erectophile, paraheliotropic combination produces a nearly uniform and low $G \approx 0.38$ across the entire hemisphere, because vertical leaves oriented perpendicular to the beam present a minimal and nearly direction-independent cross-section. Together, Figs. 5 and 6 make visible the full range of extinction geometries that the G -function encodes.

§9. The G -function in coniferous canopies.

In coniferous canopies, individual needles are small, cylindrical, and densely packed along shoots. The mutual shading of needles within a single shoot causes the effective projection area of the shoot to be substantially less than the sum of the individual needle projections. The turbid-medium formulation developed in the preceding sections, which treats each foliage element as an independent planar scatterer, is therefore inadequate at the needle level.

The resolution, introduced by Oker-Blom and Smolander (1988) and developed by Stenberg and collaborators in a series of papers, is to treat the shoot—the twig together with its complement of needles—as the basic scattering element, replacing the individual needle. The central quantity in this shoot-level formulation is the Silhouette to Total needle Area Ratio, abbreviated STAR.

For a given shoot viewed from a direction Ω , STAR is defined as

$$\text{STAR}(\Omega) = \frac{S(\Omega)}{A_t}, \quad (14)$$

where $S(\Omega)$ is the silhouette area of the shoot as seen from direction Ω , and A_t is the total (or hemi-surface, i.e., half the total surface) area of all needles on the shoot.

The spherically averaged STAR is obtained by averaging over all directions uniformly:

$$\overline{\text{STAR}} = \frac{1}{4\pi} \int_{4\pi} \text{STAR}(\Omega) d\Omega. \quad (15)$$

For a single convex body—such as an isolated cylindrical needle—the mean projection averaged over all directions equals exactly one-quarter of the total surface area. Thus, for a shoot with no within-shoot shading,

$$\overline{\text{STAR}}_{\text{no shading}} = 0.25. \quad (16)$$

Any reduction below this value is attributable to the mutual shading of needles within the shoot. Oker-Blom and Smolander (1988) measured $\overline{\text{STAR}}$ for 258 Scots pine shoots and found values ranging from 0.09 to 0.21, with a mean of 0.14—representing a 43% reduction in light interception efficiency due to within-shoot shading. In the absence of species-specific data, $\overline{\text{STAR}} \approx 0.16$ provides a reasonable approximation for a wide range of shoot structures.

§10. Concluding remarks.

We have derived the extinction coefficient for a vegetation canopy from first principles, proceeding from the probability of photon–leaf interaction in an elementary volume to the factored form $\sigma = u_L G$. The G -function encodes the geometric properties of the foliage, is independent of wavelength, and satisfies the normalization $\langle G \rangle = 1/2$ when averaged over all directions. Explicit analytical forms were given for the classical leaf-angle distributions (with complete derivations in Appendix E and normalization proofs in Appendix C), and the extension to non-uniform azimuthal distributions was treated in Appendix B. The adaptation required for coniferous canopies—replacing the leaf with the shoot and G with STAR—was presented in §9.

The extinction coefficient as derived here enters directly into the integro-differential and integral formulations of the radiative transfer equation for vegetation canopies. Its determination is

therefore the essential first step in any quantitative treatment of radiation propagation through vegetated media.

References.

1. Ross, J. 1981. *The radiation regime and architecture of plant stands*. Boston: Dr. W. Junk Publishers.
2. Oker-Blom, P., & Smolander, H. (1988). The ratio of shoot silhouette area to total needle area in Scots pine. *Forest Science*, 34(4), 894-906.

Appendix — Derivations of the G-Function

Appendix A. Definitions, Notation, and Conventions

§1. The G-Function

The geometry function $G(\mathbf{\Omega})$ describes the mean projection of unit foliage area in the direction $\mathbf{\Omega}$. It is defined as

$$G(\mathbf{\Omega}) = \frac{1}{2\pi} \int_0^{2\pi} g_L(\mathbf{\Omega}_L) |\mathbf{\Omega}_L \cdot \mathbf{\Omega}| d\mathbf{\Omega}_L \quad (\text{A.1})$$

where $\mathbf{\Omega}$ is the direction of projection (the beam direction), $\mathbf{\Omega}_L$ is the outward normal to the leaf surface, and $g_L(\mathbf{\Omega}_L)$ is the leaf-normal orientation density. The integration extends over the upper hemisphere of leaf-normal directions, with solid-angle element $d\mathbf{\Omega}_L = \sin\theta_L d\theta_L d\varphi_L$.

§2. Factorization of the Leaf-Normal Distribution

We assume that the leaf-normal orientation density factorizes into an inclination part and an azimuthal part:

$$g_L(\mathbf{\Omega}_L) = \bar{g}_L(\theta_L) \cdot h_L(\varphi_L) \quad (\text{A.2})$$

The inclination distribution $\bar{g}_L(\theta_L)$ is a probability density with respect to the solid-angle measure $\sin\theta_L d\theta_L$ on the upper hemisphere:

$$\int_0^{\pi/2} \bar{g}_L(\theta_L) \sin\theta_L d\theta_L = 1 \quad (\text{A.3})$$

The azimuthal distribution satisfies

$$\frac{1}{2\pi} \int_0^{2\pi} h_L(\varphi_L) d\varphi_L = 1 \quad (\text{A.4})$$

§3. Notation

Throughout we write $\mu = \cos\theta$ for the direction cosine of the beam, $\mu_L = \cos\theta_L$ for that of the leaf normal, θ and φ for the polar and azimuthal angles of the beam $\mathbf{\Omega}$, and θ_L and φ_L for those of the leaf normal $\mathbf{\Omega}_L$.

All angles are measured from the outward normal to the horizontal plane, so $\theta \in [0, \pi/2]$ and $\theta_L \in [0, \pi/2]$.

The dot product of the two unit vectors is

$$\mathbf{\Omega}_L \cdot \mathbf{\Omega} = \cos\theta_L \cos\theta + \sin\theta_L \sin\theta \cos(\varphi_L - \varphi) \quad (\text{A.5})$$

We frequently use the abbreviations

$$a \equiv \mu\mu_L = \cos\theta\cos\theta_L, \quad b \equiv \sin\theta\sin\theta_L \quad (\text{A.6})$$

§4. The Normalization Condition for G

The G-function satisfies the normalization condition

$$\frac{1}{2\pi} \int_{2\pi} G(\boldsymbol{\Omega}) d\boldsymbol{\Omega} = \frac{1}{2} \quad (\text{A.7})$$

When G depends on both θ and φ , this takes the form

$$\frac{1}{2\pi} \int_0^{2\pi} \int_0^{\pi/2} G(\theta, \varphi) \sin\theta d\theta d\varphi = \frac{1}{2} \quad (\text{A.8})$$

When G is independent of φ (as occurs when $h_L = 1$), this reduces to

$$\int_0^{\pi/2} G(\theta) \sin\theta d\theta = \frac{1}{2} \quad (\text{A.9})$$

§5. Models for the Inclination Distribution

Six models for $\bar{g}_L(\theta_L)$ are considered:

Constant (single inclination angle): $\bar{g}_L(\theta_L) = \delta(\theta_L - \theta_0)/\sin\theta_0$, with special cases $\theta_0 = 0$ (horizontal) and $\theta_0 = \pi/2$ (vertical).

Uniform (isotropic): $\bar{g}_L(\theta_L) = 1$.

Planophile (predominantly horizontal): $\bar{g}_L(\theta_L) = 3\cos^2\theta_L$.

Erectophile (predominantly vertical): $\bar{g}_L(\theta_L) = \frac{3}{2}\sin^2\theta_L$.

Plagiophile (predominantly at 45 degrees): $\bar{g}_L(\theta_L) = \frac{15}{8}\sin^2 2\theta_L$.

Extremophile (bimodal, horizontal and vertical): $\bar{g}_L(\theta_L) = \frac{15}{7}\cos^2 2\theta_L$.

§6. The Azimuthal Distribution Model

The azimuthal distribution of leaf normals is given by

$$h_L(\varphi_L, \varphi) = 2\cos^2(\varphi - \varphi_L - \eta) \quad (\text{A.10})$$

The parameter η controls the preferred leaf azimuthal orientation. Three special cases are of interest: Diaheliotropic ($\eta = 0$, leaves preferentially face the beam), Paraheliotropic ($\eta = \pi/2$, leaves preferentially face perpendicular to the beam), and Uniform ($h_L = 1$, no azimuthal preference). The uniform case is recovered by setting $\Delta\psi = 0$ in the general framework developed below, or equivalently by averaging h_L over η .

Normalization. Setting $\alpha = \varphi - \varphi_L - \eta$ and applying $\cos^2 \alpha = \frac{1}{2}(1 + \cos 2\alpha)$:

$$\frac{1}{2\pi} \int_0^{2\pi} 2 \cos^2 \alpha \, d\varphi_L = \frac{1}{2\pi} \int_0^{2\pi} (1 + \cos 2\alpha) \, d\varphi_L = \frac{1}{2\pi} [2\pi + 0] = 1 \quad \checkmark$$

Appendix B. The Azimuthal Projection Integral

§7. Formulation

With the non-uniform azimuthal distribution Eq. (A.10), the azimuthal projection integral is

$$\psi_{NU}(\mu, \mu_L, \eta) = \frac{1}{2\pi} \int_0^{2\pi} h_L(\varphi_L, \varphi) |\mathbf{\Omega}_L \cdot \mathbf{\Omega}| d\varphi_L \quad (\text{B.1})$$

Substituting h_L and the dot product, and introducing $\alpha = \varphi_L - \varphi$:

$$\psi_{NU} = \frac{1}{\pi} \int_0^{2\pi} \cos^2(\alpha + \eta) |a + b\cos\alpha| d\alpha \quad (\text{B.2})$$

§8. Decomposition into Uniform Part and Correction

Applying $\cos^2(\alpha + \eta) = \frac{1}{2} + \frac{1}{2} \cos(2\alpha + 2\eta)$ and distributing:

$$\psi_{NU} = \underbrace{\frac{1}{2\pi} \int_0^{2\pi} |a + b\cos\alpha| d\alpha}_{\check{\psi}_U} + \underbrace{\frac{1}{2\pi} \int_0^{2\pi} \cos(2\alpha + 2\eta) |a + b\cos\alpha| d\alpha}_{\Delta\check{\psi}}$$

This yields the fundamental decomposition

$$\psi_{NU}(\mu, \mu_L, \eta) = \psi_U(\mu, \mu_L) + \Delta\psi(\mu, \mu_L, \eta) \quad (\text{B.3})$$

where ψ_U is the uniform- h_L projection integral and $\Delta\psi$ is the correction arising from azimuthal anisotropy. Setting $h_L = 1$ (uniform case) corresponds to $\Delta\psi = 0$.

§9. Evaluation of the Uniform Part

Case 1: $|a| \geq b$. The expression $a + b\cos\alpha$ does not change sign over $[0, 2\pi]$:

$$\psi_U = \frac{1}{2\pi} \left| \int_0^{2\pi} (a + b\cos\alpha) d\alpha \right| = \frac{1}{2\pi} |2\pi a| = |\mu\mu_L|$$

Case 2: $|a| < b$. The expression changes sign at $\alpha = \varphi_t$ and $\alpha = 2\pi - \varphi_t$, where

$$\varphi_t \equiv \arccos\left(-\frac{\mu\mu_L}{\sin\theta\sin\theta_L}\right) \quad (\text{B.4})$$

Splitting the integral at these zeros, using the symmetry $\alpha \rightarrow 2\pi - \alpha$ to combine the first and third sub-intervals, and noting that $I_+ + I_- = 2\pi a$ where I_+ and I_- are the integrals over the positive and negative regions:

$$2\pi\psi_U = 2I_+ - 2\pi a, \quad I_+ = 2(a\varphi_t + b\sin\varphi_t)$$

Simplifying:

$$\psi_U = \mu\mu_L \left(\frac{2\varphi_t}{\pi} - 1 \right) + \frac{2}{\pi} \sin\theta \sin\theta_L \sin\varphi_t \quad (\text{B.5})$$

Summary. The uniform projection integral is:

When $|\mu\mu_L| \geq \sin\theta \sin\theta_L$:

$$\psi_U = |\mu\mu_L| \quad (\text{B.6a})$$

When $|\mu\mu_L| < \sin\theta \sin\theta_L$:

$$\psi_U = \mu\mu_L \left(\frac{2\varphi_t}{\pi} - 1 \right) + \frac{2}{\pi} \sin\theta \sin\theta_L \sin\varphi_t \quad (\text{B.6b})$$

§10. Evaluation of the Correction

Case I: $|a| \geq b$. Expanding $\cos(2\alpha + 2\eta)(a + b\cos\alpha)$ and integrating term by term: the first term vanishes because $\cos(2\alpha + 2\eta)$ has zero mean, and the second vanishes by the product-to-sum formula $\cos(2\alpha + 2\eta)\cos\alpha = \frac{1}{2}[\cos(\alpha + 2\eta) + \cos(3\alpha + 2\eta)]$, both of which integrate to zero over $[0, 2\pi]$. Therefore

$$\Delta\psi = 0 \quad (\text{Case I: } |a| \geq b) \quad (\text{B.7})$$

This means the non-uniform azimuthal preference has no effect when $|\mu\mu_L| \geq \sin\theta \sin\theta_L$.

Case II: $|a| < b$. Resolving the absolute value, splitting at φ_t and $2\pi - \varphi_t$, using the antiderivatives $\int \cos(2\alpha + 2\eta) d\alpha = \frac{1}{2} \sin(2\alpha + 2\eta)$ and $\int \cos(2\alpha + 2\eta)\cos\alpha d\alpha = \frac{1}{2} \sin(\alpha + 2\eta) + \frac{1}{6} \sin(3\alpha + 2\eta)$, and evaluating at the boundaries:

$$\Delta\psi = \frac{a}{2\pi} [2\sin(2\varphi_t + 2\eta) - \sin 4\eta] + \frac{b}{2\pi} \left[\sin(\varphi_t + 2\eta) + \frac{1}{3} \sin(3\varphi_t + 2\eta) - \frac{4}{3} \sin 2\eta \right] \quad (\text{B.8})$$

§11. Summary

The non-uniform azimuthal projection integral is $\psi_{NU} = \psi_U + \Delta\psi$ (Eq. (B.3)), where ψ_U is given by Eqs. (B.6a) and (B.6b), and $\Delta\psi = 0$ when $|a| \geq b$ (Eq. (B.7)) while $\Delta\psi$ is given by Eq. (B.8) otherwise. The uniform case ($h_L = 1$) is recovered by setting $\Delta\psi = 0$ throughout.

Appendix C. The Master Normalization Identity

§12. Theorem

For any fixed $\theta_L \in [0, \pi/2]$ and any azimuthal distribution h_L satisfying Eq. (A.4),

$$\frac{1}{2\pi} \int_0^{2\pi} \int_0^{\pi/2} \psi_{NU}(\mu, \mu_L, \eta) \sin\theta \, d\theta \, d\varphi = \frac{1}{2} \quad (\text{C.1})$$

§13. Proof

By Fubini's theorem, the left side equals $\frac{1}{2\pi} \int h_L(\varphi_L) \left[\frac{1}{2\pi} \int \int |\mathbf{\Omega}_L \cdot \mathbf{\Omega}| \sin\theta \, d\theta \, d\varphi \right] d\varphi_L$. The bracketed quantity is the hemisphere-average of $|\mathbf{\Omega}_L \cdot \mathbf{\Omega}|$ over all beam directions. Aligning the polar axis with $\mathbf{\Omega}_L$, this becomes $\int_0^{\pi/2} \cos\alpha \sin\alpha \, d\alpha = 1/2$, independent of $\mathbf{\Omega}_L$. The result then follows from the normalization of h_L (Eq. (A.4)).

§14. Consequence for the G-Function

The G-function is $G(\theta, \varphi) = \int_0^{\pi/2} \bar{g}_L(\theta_L) \psi_{NU} \sin\theta_L \, d\theta_L$. Integrating over beam directions and using Fubini:

$$\frac{1}{2\pi} \int_0^{2\pi} \int_0^{\pi/2} G \sin\theta \, d\theta \, d\varphi = \frac{1}{2} \int_0^{\pi/2} \bar{g}_L \sin\theta_L \, d\theta_L$$

The normalization of G (Eq. (A.8)) therefore reduces to the normalization of $\bar{g}_L$ (Eq. (A.3)). Since every inclination distribution in §5 satisfies this, normalization is guaranteed in all cases below.

Appendix D. The Fundamental Integrals

§15. Motivation

The four trigonometric inclination distributions expand as $c_0 + c_2 \cos 2\theta_L + c_4 \cos 4\theta_L$ via double-angle identities. The uniform- h_L G-function then decomposes as

$$G_U = c_0 \mathcal{J}_0 + c_2 \mathcal{J}_2 + c_4 \mathcal{J}_4 \quad (\text{D.1})$$

where $\mathcal{J}_n \equiv \int_0^{\pi/2} \cos(n\theta_L) \psi_U(\mu, \mu_L) \sin\theta_L d\theta_L$.

§16. Results

By the coordinate-alignment argument:

$$\mathcal{J}_0 = \frac{1}{2} \quad (\text{D.2})$$

By splitting the integration domain at $\theta_L^* = \pi/2 - \theta$ and evaluating each branch:

$$\mathcal{J}_2 = -\frac{\sin^2 \theta}{4} \quad (\text{D.3})$$

By an analogous calculation:

$$\mathcal{J}_4 = -\frac{\cos^4 \theta}{6} \quad (\text{D.4})$$

Appendix E. G-Functions for All Leaf Inclination Models

Problem 1. Constant Leaf Inclination

Problem Statement

The leaf inclination distribution is concentrated at a single angle θ_0 :

$$\bar{g}_L(\theta_L) = \frac{\delta(\theta_L - \theta_0)}{\sin\theta_0} \quad (\text{E.1})$$

Normalization: $\int_0^{\pi/2} [\delta(\theta_L - \theta_0)/\sin\theta_0] \sin\theta_L d\theta_L = \sin\theta_0/\sin\theta_0 = 1$. ✓

Derivation

By the sifting property of the delta function:

$$G(\theta, \eta) = \psi_{NU}(\mu, \cos\theta, \eta) \quad (\text{E.2})$$

With $a = \cos\theta\cos\theta_0$ and $b = \sin\theta\sin\theta_0$, the branch condition $|a| \geq b$ is equivalent to $\theta + \theta_0 \leq \pi/2$.

When $\theta + \theta_0 \leq \pi/2$: $\Delta\psi = 0$ and $\psi_U = \cos\theta\cos\theta_0$, so $G = \cos\theta\cos\theta_0$.

When $\theta + \theta_0 > \pi/2$: writing $\varphi_t = \arccos(-\cot\theta\cot\theta_0)$,

$$\begin{aligned} G = & \cos\theta\cos\theta_0 \left(\frac{2\varphi_t}{\pi} - 1 \right) + \frac{2}{\pi} \sin\theta\sin\theta_0 \sin\varphi_t \\ & + \frac{\cos\theta\cos\theta_0}{2\pi} [2\sin(2\varphi_t + 2\eta) - \sin 4\eta] \\ & + \frac{\sin\theta\sin\theta_0}{2\pi} \left[\sin(\varphi_t + 2\eta) + \frac{1}{3} \sin(3\varphi_t + 2\eta) - \frac{4}{3} \sin 2\eta \right] \end{aligned} \quad (\text{E.3})$$

Normalization

Since $\bar{g}_L$ is properly normalized, the master identity (§14) guarantees Eq. (A.8). ✓

Special Cases

Diaheliotropic ($\eta = 0$). Setting $\eta = 0$: $\sin 4\eta = 0$, $\sin 2\eta = 0$. When $\theta + \theta_0 \leq \pi/2$: $G = \cos\theta\cos\theta_0$. When $\theta + \theta_0 > \pi/2$, applying $\sin 2\varphi_t = 2\sin\varphi_t\cos\varphi_t$ and $\sin 3\varphi_t = 3\sin\varphi_t - 4\sin^3\varphi_t$:

$$\Delta\psi|_{\eta=0} = \frac{2a\sin\varphi_t\cos\varphi_t}{\pi} + \frac{b\sin\varphi_t}{\pi} \left[1 - \frac{2}{3} \sin^2\varphi_t \right] \quad (\text{E.4})$$

Paraheliotropic ($\eta = \pi/2$). Setting $\eta = \pi/2$: $\sin(2\varphi_t + \pi) = -\sin 2\varphi_t$, $\sin(\varphi_t + \pi) = -\sin\varphi_t$, $\sin(3\varphi_t + \pi) = -\sin 3\varphi_t$. Therefore

$$\Delta\psi|_{\eta=\pi/2} = -\Delta\psi|_{\eta=0} \quad (\text{E.5})$$

The paraheliotropic correction is the exact negative of the diaheliotropic correction.

Uniform ($h_L = 1$). Setting $\Delta\psi = 0$: $G = \psi_U(\mu, \cos\theta_0)$, the piecewise expression of Eqs. (B.6a) and (B.6b) at $\mu_L = \cos\theta_0$.

Sub-case: Vertical Leaves

With $\theta_0 = \pi/2$: $a = 0$, $b = \sin\theta$, $\varphi_t = \pi/2$. The uniform part is $\psi_U = (2/\pi)\sin\theta$. The correction becomes:

$$\Delta\psi = \frac{\sin\theta}{3\pi} [\cos 2\eta - 2\sin 2\eta] \quad (\text{E.6})$$

so that $G = (2/\pi)\sin\theta + (\sin\theta/3\pi)(\cos 2\eta - 2\sin 2\eta)$ for $\theta > 0$. The special cases are: diaheliotropic $G = (7/3\pi)\sin\theta$, paraheliotropic $G = (5/3\pi)\sin\theta$, uniform $G = (2/\pi)\sin\theta$. Note $G_{\text{dia}} > G_{\text{uniform}} > G_{\text{para}}$ for all $\theta > 0$.

Sub-case: Horizontal Leaves

With $\theta_0 = 0$: $a = \cos\theta$, $b = 0$. The condition $|a| \geq b$ is always satisfied, so $\Delta\psi = 0$ and

$$G = \cos\theta \quad (\text{E.7})$$

regardless of η . The azimuthal orientation is irrelevant for horizontal leaves.

Problem 2. Uniform Leaf Inclination

Problem Statement

$$\bar{g}_L(\theta_L) = 1 \quad (\text{E.8})$$

Normalization: $\int_0^{\pi/2} \sin \theta_L d\theta_L = [-\cos \theta_L]_0^{\pi/2} = 1. \checkmark$

Derivation

$$G(\theta, \eta) = \int_0^{\pi/2} \psi_{NU} \sin \theta_L d\theta_L = \int_0^{\pi/2} \psi_U \sin \theta_L d\theta_L + \int_0^{\pi/2} \Delta \psi \sin \theta_L d\theta_L$$

$\mathcal{I}_0 \stackrel{\sim}{=} 1/2$

Since $\Delta \psi = 0$ for $\theta_L \leq \pi/2 - \theta$:

$$G(\theta, \eta) = \frac{1}{2} + \int_{\pi/2-\theta}^{\pi/2} \Delta \psi(\mu, \mu_L, \eta) \sin \theta_L d\theta_L \quad (\text{E.9})$$

Normalization

By §14, with $\int \bar{g}_L \sin \theta_L d\theta_L = 1. \checkmark$

Special Cases

Diaheliotropic ($\eta = 0$). $G|_{\eta=0} = 1/2 + \int_{\pi/2-\theta}^{\pi/2} \Delta \psi|_{\eta=0} \sin \theta_L d\theta_L$, with $\Delta \psi|_{\eta=0}$ from Eq. (E.4).

Paraheliotropic ($\eta = \pi/2$). By Eq. (E.5):

$$G|_{\eta=\pi/2} = 1 - G|_{\eta=0} \quad (\text{E.10})$$

The paraheliotropic and diaheliotropic G-functions are symmetric about $G = 1/2$.

Uniform ($h_L = 1$). $\Delta \psi = 0$ gives $G = 1/2$, independent of θ .

Problem 3. Planophile Leaf Inclination

Problem Statement

$$\bar{g}_L(\theta_L) = 3\cos^2\theta_L \quad (\text{E.11})$$

Normalization: $3 \int_0^1 u^2 du = 3 \cdot 1/3 = 1. \checkmark$

Derivation

Using $3\cos^2\theta_L = \frac{3}{2}(1 + \cos 2\theta_L)$, the coefficients are $c_0 = 3/2$, $c_2 = 3/2$, $c_4 = 0$. The uniform part is $G_U = \frac{3}{2} \cdot \frac{1}{2} + \frac{3}{2} \cdot (-\sin^2\theta/4) = \frac{3}{4} - \frac{3\sin^2\theta}{8}$. Writing $\sin^2\theta = 1 - \cos^2\theta$:

$$G_U = \frac{3(1+\cos^2\theta)}{8} \quad (\text{E.12})$$

The correction is

$$\Delta G(\theta, \eta) = 3 \int_{\pi/2-\theta}^{\pi/2} \cos^2 \theta_L \Delta\psi(\mu, \mu_L, \eta) \sin\theta_L d\theta_L \quad (\text{E.13})$$

The complete G-function is $G = G_U + \Delta G$.

Normalization

By §14, with $\int \bar{g}_L \sin\theta_L d\theta_L = 1. \checkmark$

Direct verification for the uniform case: $(3/8) \int_0^1 (1 + u^2) du = (3/8)(4/3) = 1/2. \checkmark$

Special Cases

Diaheliotropic ($\eta = 0$). $G = 3(1 + \cos^2\theta)/8 + \Delta G|_{\eta=0}$.

Paraheliotropic ($\eta = \pi/2$). By Eq. (E.5): $G = 3(1 + \cos^2\theta)/8 - \Delta G|_{\eta=0}$.

Uniform ($h_L = 1$). $G = 3(1 + \cos^2\theta)/8$.

Problem 4. Erectophile Leaf Inclination

Problem Statement

$$\bar{g}_L(\theta_L) = \frac{3}{2} \sin^2 \theta_L \quad (\text{E.14})$$

Normalization: $\frac{3}{2} \int_0^1 (1 - u^2) du = \frac{3}{2} \cdot 2/3 = 1. \checkmark$

Derivation

Using $\frac{3}{2} \sin^2 \theta_L = \frac{3}{4} (1 - \cos 2\theta_L)$, the coefficients are $c_0 = 3/4$, $c_2 = -3/4$, $c_4 = 0$. The uniform part is:

$$G_U = \frac{3(2 + \sin^2 \theta)}{16} \quad (\text{E.15})$$

The correction is $\Delta G(\theta, \eta) = \frac{3}{2} \int_{\pi/2 - \theta}^{\pi/2} \sin^2 \theta_L \Delta \psi \sin \theta_L d\theta_L$, and $G = G_U + \Delta G$.

Normalization

By §14. Direct verification for the uniform case: $(3/16) \int_0^1 (3 - u^2) du = (3/16)(8/3) = 1/2. \checkmark$

Special Cases

Diaheliotropic ($\eta = 0$). $G = 3(2 + \sin^2 \theta)/16 + \Delta G|_{\eta=0}$.

Paraheliotropic ($\eta = \pi/2$). $G = 3(2 + \sin^2 \theta)/16 - \Delta G|_{\eta=0}$.

Uniform ($h_L = 1$). $G = 3(2 + \sin^2 \theta)/16$.

Problem 5. Plagiophile Leaf Inclination

Problem Statement

$$\bar{g}_L(\theta_L) = \frac{15}{8} \sin^2 2\theta_L \quad (\text{E.16})$$

Normalization: $\frac{15}{2} \int_0^1 (1 - u^2) u^2 du = \frac{15}{2} (1/3 - 1/5) = \frac{15}{2} \cdot 2/15 = 1. \checkmark$

Derivation

Using $\frac{15}{8} \sin^2 2\theta_L = \frac{15}{16} (1 - \cos 4\theta_L)$, the coefficients are $c_0 = 15/16$, $c_2 = 0$, $c_4 = -15/16$.

The uniform part is $G_U = \frac{15}{32} + \frac{15 \cos^4 \theta}{96}$:

$$G_U = \frac{5(3 + \cos^4 \theta)}{32} \quad (\text{E.17})$$

The correction is $\Delta G(\theta, \eta) = \frac{15}{8} \int_{\pi/2 - \theta}^{\pi/2} \sin^2 2\theta_L \Delta\psi \sin \theta_L d\theta_L$, and $G = G_U + \Delta G$.

Normalization

By §14. Direct verification: $(5/32) \int_0^1 (3 + u^4) du = (5/32)(16/5) = 1/2. \checkmark$

Special Cases

Diaheliotropic ($\eta = 0$). $G = 5(3 + \cos^4 \theta)/32 + \Delta G|_{\eta=0}$.

Paraheliotropic ($\eta = \pi/2$). $G = 5(3 + \cos^4 \theta)/32 - \Delta G|_{\eta=0}$.

Uniform ($h_L = 1$). $G = 5(3 + \cos^4 \theta)/32$.

Problem 6. Extremophile Leaf Inclination

Problem Statement

$$\bar{g}_L(\theta_L) = \frac{15}{7} \cos^2 2\theta_L \quad (\text{E.18})$$

Normalization: $\frac{15}{7} \int_0^1 (4u^4 - 4u^2 + 1) du = \frac{15}{7} (4/5 - 4/3 + 1) = \frac{15}{7} \cdot 7/15 = 1. \checkmark$

Derivation

Using $\frac{15}{7} \cos^2 2\theta_L = \frac{15}{14} (1 + \cos 4\theta_L)$, the coefficients are $c_0 = 15/14$, $c_2 = 0$, $c_4 = 15/14$. The uniform part is $G_U = \frac{15}{28} - \frac{15 \cos^4 \theta}{84}$:

$$G_U = \frac{5(3 - \cos^4 \theta)}{28} \quad (\text{E.19})$$

The correction is $\Delta G(\theta, \eta) = \frac{15}{7} \int_{\pi/2 - \theta}^{\pi/2} \cos^2 2\theta_L \Delta\psi \sin \theta_L d\theta_L$, and $G = G_U + \Delta G$.

Normalization

By §14. Direct verification: $(5/28) \int_0^1 (3 - u^4) du = (5/28)(14/5) = 1/2. \checkmark$

Special Cases

Diaheliotropic ($\eta = 0$). $G = 5(3 - \cos^4 \theta)/28 + \Delta G|_{\eta=0}$.

Paraheliotropic ($\eta = \pi/2$). $G = 5(3 - \cos^4 \theta)/28 - \Delta G|_{\eta=0}$.

Uniform ($h_L = 1$). $G = 5(3 - \cos^4 \theta)/28$.

Appendix F. Summary of Results

§17. Structural Decomposition

For every combination of inclination model $\bar{g}_L$ and the non-uniform azimuthal model $h_L = 2\cos^2(\varphi - \varphi_L - \eta)$, the G-function takes the universal form

$$G(\theta, \eta) = G_U(\theta) + \Delta G(\theta, \eta) \quad (\text{F.1})$$

where G_U is the uniform- h_L result and ΔG is the correction from azimuthal anisotropy.

§18. Comprehensive Table of G-Functions

The table below collects results for all six inclination models. For each model, the general G-function under the non-uniform azimuthal distribution is given, followed by its three special cases.

Inclination Model	Azimuthal Case	G-Function	Eq.
Constant θ_0	General	$G = \psi_U + \Delta\psi$	(E.3)
	Diaheliotropic	$G = \psi_U + \Delta\psi _{\eta=0}$	(E.4)
	Paraheliotropic	$G = \psi_U - \Delta\psi _{\eta=0}$	(E.5)
	Uniform h_l	Piecewise: Eqs. (B.6a), (B.6b)	
Horizontal ($\theta_0=0$)	All cases	$G = \cos\theta$	(E.7)
Vertical ($\theta_0=\pi/2$)	General	$G = (2/\pi)\sin\theta + (\sin\theta/3\pi)(\cos 2\eta - 2\sin 2\eta)$	(E.6)
	Diaheliotropic	$G = (7/3\pi)\sin\theta$	
	Paraheliotropic	$G = (5/3\pi)\sin\theta$	
	Uniform h_l	$G = (2/\pi)\sin\theta$	
Uniform	General	$G = 1/2 + \int \Delta\psi \sin\theta_l d\theta_l$	(E.9)
	Diaheliotropic	$G = 1/2 + \text{correction}$	
	Paraheliotropic	$G = 1 - G _{\eta=0}$	(E.10)
	Uniform h_l	$G = 1/2$	
Planophile	General	$G = 3(1+\cos^2\theta)/8 + \Delta G$	(E.12)
	Diaheliotropic	$G = 3(1+\cos^2\theta)/8 + \Delta G _{\eta=0}$	
	Paraheliotropic	$G = 3(1+\cos^2\theta)/8 - \Delta G _{\eta=0}$	
	Uniform h_l	$G = 3(1+\cos^2\theta)/8$	
Erectophile	General	$G = 3(2+\sin^2\theta)/16 + \Delta G$	(E.15)
	Diaheliotropic	$G = 3(2+\sin^2\theta)/16 + \Delta G _{\eta=0}$	
	Paraheliotropic	$G = 3(2+\sin^2\theta)/16 - \Delta G _{\eta=0}$	
	Uniform h_l	$G = 3(2+\sin^2\theta)/16$	
Plagiophile	General	$G = 5(3+\cos^4\theta)/32 + \Delta G$	(E.17)

Inclination Model	Azimuthal Case	G-Function	Eq.
	Diaheliotropic	$G = 5(3+\cos^4\theta)/32 + \Delta G _{\eta=0}$	
	Paraheliotropic	$G = 5(3+\cos^4\theta)/32 - \Delta G _{\eta=0}$	
	Uniform h_l	$G = 5(3+\cos^4\theta)/32$	
Extremophile	General	$G = 5(3-\cos^4\theta)/28 + \Delta G$	(E.19)
	Diaheliotropic	$G = 5(3-\cos^4\theta)/28 + \Delta G _{\eta=0}$	
	Paraheliotropic	$G = 5(3-\cos^4\theta)/28 - \Delta G _{\eta=0}$	
	Uniform h_l	$G = 5(3-\cos^4\theta)/28$	

§19. Key Structural Properties

3. **Diaheliotropic-Paraheliotropic symmetry.** $\Delta G|_{\eta=\pi/2} = -\Delta G|_{\eta=0}$ for all inclination models (Eq. (E.5)).
4. **Horizontal leaves are immune to azimuthal anisotropy.** When $\theta_0 = 0$, $b = 0$ and $|a| \geq b$ always, so $\Delta\psi = 0$ and $G = \cos\theta$ regardless of η .
5. **Recovery of the uniform case.** Setting $h_L = 1$ gives $\Delta\psi = 0$ and $G = G_U$ for every model.
6. **Universal normalization.** The master identity Eq. (C.1), combined with the normalization of $\bar{g}_L$ and Fubini's theorem (§14), guarantees Eq. (A.7) for every combination treated in this appendix.